

■ FarmsKing Technical Trainer System - Admin Architecture & Operational Guide

■ Executive Overview

FarmsKing Technical Trainer System is a state & district level farmer onboarding, app guidance, and technical support hierarchy. It ensures that every newly registered farmer gets direct phone & WhatsApp training assistance from a designated Technical Trainer.

■■ 1. Technical Architecture & Data Storage (■■■■■ ■■ ■■■■ ■■)

■■ Backend Database Schema (backend/prisma/schema.prisma)

- **User Role:** TECHNICAL_TRAINER added to Role enum.
- **TrainerAssignment Model:**
 - **trainerId:** Linked to trainer's User record.
 - **level:** BASELINE (Level 1 Part-time) or UPLINE (Level 2 Senior).
 - **grade:** GRADE_A (4.5-5.0★), GRADE_B (3.5-4.4★), GRADE_C (<3.5★).
 - **state & district:** Geographic assignment zone.
 - **isAvailable, availableFrom, availableTo, shiftType:** Duty status & available call windows.
 - **commissionRate:** Custom wallet reward setting per trainer.
- **FarmerTrainingLog Model:**
 - **farmerId:** Unique farmer assigned.
 - **trainerId:** Baseline Technical Trainer assigned.
 - **isCallRequested, preferredCallSlot:** Farmer's 1-tap call request data (MORNING_9_12, AFTERNOON_12_4, EVENING_4_7, ANYTIME).
 - **isForwarded, forwardedToId, forwardReason:** Baseline to Upline escalation record.
 - **status:** PENDING_CALL, IN_PROGRESS, WAITING_VERIFICATION, VERIFIED_AND_PAID, UNREACHABLE.
 - **rating:** 1 to 5 Star Rating given by Farmer.
 - **payoutAmount & payoutTxId:** Instant FarmsKing Wallet Credit transaction ID.

■ 2. Backend Modules & API Endpoints

■ Files:

- backend/src/modules/trainers/trainers.module.ts
- backend/src/modules/trainers/trainers.service.ts
- backend/src/modules/trainers/trainers.controller.ts

■ Key API Endpoints:

- GET /api/v1/trainers/my-assigned-farmers -> Technical Trainer fetches assigned farmers list.
- POST /api/v1/trainers/request-call -> Farmer requests 1-tap callback with preferred time slot.
- POST /api/v1/trainers/farmer-verify -> Farmer submits 1-5★ Star Rating & releases wallet reward to trainer.
- POST /api/v1/trainers/forward-upline -> Baseline Trainer escalates complex issue to Upline Senior.
- POST /api/v1/trainers/update-availability -> Trainer toggles Online/Offline or shift hours.
- POST /api/v1/trainers/assignments -> Admin assigns trainer to state/district using King ID.
- GET /api/v1/trainers/admin-reports -> Admin live accountability dashboard & call logs.

■ 3. Frontend Mobile & Web App Screens

■ Files:

- frontend/app/(tabs)/trainer-dashboard.tsx -> Trainer Mobile Dashboard with Call, WhatsApp, and Forward buttons.
- frontend/src/components/FarmerTrainingRatingBanner.tsx -> Farmer 1-to-5 Star Rating & Feedback Banner on Farmer Home Screen.
- frontend/src/api/trainers.api.ts -> Frontend API client functions.

■ 4. 3-Tier Reward & Wallet System Rules

- Star Rating | Trainer Wallet Reward | Performance Grade Impact
- :--- | :--- | :---
- 5★ (■■■■■■■■) | ■25 Credited Instant | Grade A (Top Trainer - Upline Eligible)
- 4★ (■■■■■) | ■15 Credited Instant | Grade B (Regular Trainer)
- 1-3★ (■■■■) | ■0 Credited | Grade C (Needs Improvement / Warning)

■ 5. Complete Step-by-Step Workflow (■■■■■■ ■■■■■■)

- **Farmer Registration:** When a new farmer registers in Punjab/Haryana/etc., the backend automatically checks TrainerAssignment for their State/District and assigns them to the Baseline Technical Trainer.
- **Trainer Notification & Dashboard:** The assigned Baseline Trainer sees the new farmer in their Trainer Dashboard with 1-click ■ Call and ■ WhatsApp Welcome options.
- **Farmer Call Request:** If farmer needs assistance, they tap ■ ■■■■■ ■■■■■■ ■■ ■■■ ■■■■■■ ■■ on their home screen. Trainer gets an immediate WhatsApp notification.
- **Escalation (Forwarding):** If Baseline Trainer faces a complex technical/UPI problem, they click ■ Forward to Upline, sending the case to the State Upline Senior Trainer.
- **Rating & Wallet Payout:** Once training is done, farmer rates 4★ or 5★ on their app screen. The exact reward (■15 or ■25) is instantly transferred to the Technical Trainer's FarmsKing Wallet!